Paul Thomas

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Education

CESI Graduate School of Engineering — Strasbourg, France
Master of Science in Computer Science & Engineering (Co-op Track) — *Sep. 2023 – Oct. 2028*
- **Focus**: Artificial Intelligence, Machine Learning, Embedded Systems & Distributed Computing.

Lycée Marie Curie — Strasbourg, France
French Baccalaureate with Honors (Mathematics & Computer Science) — *June 2023*

Experience

Apprentice Research Engineer — Oct. 2025 – Oct. 2028
CESI LINEACT Research Laboratory (UR 7527) — *Strasbourg, France*
- Engineered a distributed Federated Learning platform across a 20-node cluster (Raspberry Pi & Jetson).
- Integrated state-of-the-art FL algorithms into an asynchronous WebSocket and MQTT pipeline.
- Benchmarked Edge AI convergence and straggler resilience under constrained network conditions.

R&D Intern – Federated Learning — Apr. 2025 – June 2025
CESI LINEACT Research Laboratory (UR 7527) — *Strasbourg, France*
- Prototyped a distributed ML architecture to evaluate decentralized model training on resource-constrained hardware.

Founder & Full-Stack Developer — 2021 – 2024
Studio Osalys — *Strasbourg, France*
- Delivered 15+ full-stack web applications, SaaS platforms, and REST APIs using TypeScript, React, Node.js, and SQL.

Early Internships — 2019 – 2021
- **Pixacare** (AI MedTech, 2021) | **Elixir Health** (AI Health, 2021) | **neverMind** (Embedded Systems, 2019)

Projects

Axiom Chess: Spatial Computing & Vision Ecosystem | *PyTorch, Computer Vision, OpenCV, YOLO* — 2025 – 2026
- Axiom Vision: Real-time chessboard detection and piece tracking using OpenCV and deep learning pipelines.
- Axiom Light: Augmented reality projection overlay mapping dynamic moves directly onto the physical board.
- Axiom Engine: Custom Alpha-Beta search engine integrated with a Deep Neural Network for static board evaluation.

Variational Quantum Classifiers (QML) | *PennyLane, PyTorch, Qiskit* — 2026
- Engineered hybrid quantum-classical networks (VQC) and benchmarked noise robustness on complex data.

Custom Autonomous Search Robot | *ROS 2, Python, YOLO, OpenCV, SLAM, Embedded* — 2026
- Designed and built a custom autonomous mobile robot using SLAM and YOLO to map indoor spaces and locate targets.

Vehicle Route & Fleet Optimization | *Python, Graph Theory, Combinatorics, Pandas* — 2025
- Implemented operations research algorithms solving Vehicle Routing Problems (VRP) with complex time-window constraints.

Resilient Embedded Weather Station | *C++, ESP32 / Arduino, RTOS, Sensors* — 2024
- Engineered an autonomous low-power embedded sensor payload for environmental telemetry and hazard monitoring.

Honors & Certifications

3rd Prize & Special Award (Jury's Favorite): Aquathon 2025 (Regenerative Hydrology project *Hydro Regen*).

Top Scorer: AXA Challenge "Best Computer Science Student in France" (Cyber, AI/ML, Cloud & Dev) – 2025.

TOEIC Listening & Reading: 875/990 (Advanced Professional English Proficiency) – 2025.

Technical Skills

Languages: Python, C/C++, C#, TypeScript, JavaScript, SQL, Java, PHP
AI & Machine Learning: PyTorch, TensorFlow, Federated Learning, OpenCV, Scikit-Learn, TinyML, PennyLane, Qiskit
Systems & Embedded: Linux IPC, RTOS, ESP32, Arduino, Raspberry Pi, ROS/ROS2, MQTT, Sensor Fusion, Gazebo
Tools & Full-Stack: Docker, Docker Swarm, Git, CI/CD, Jupyter Notebook, React, Vue, Node.js, FastAPI, PostgreSQL
Languages (Spoken): French (Native), English (Fluent / TOEIC 875), German (Working proficiency)